

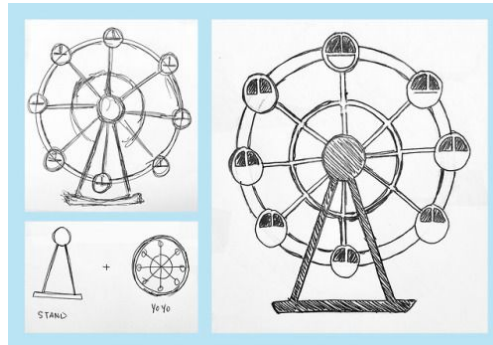
Wonderbread Collectors

Final Report

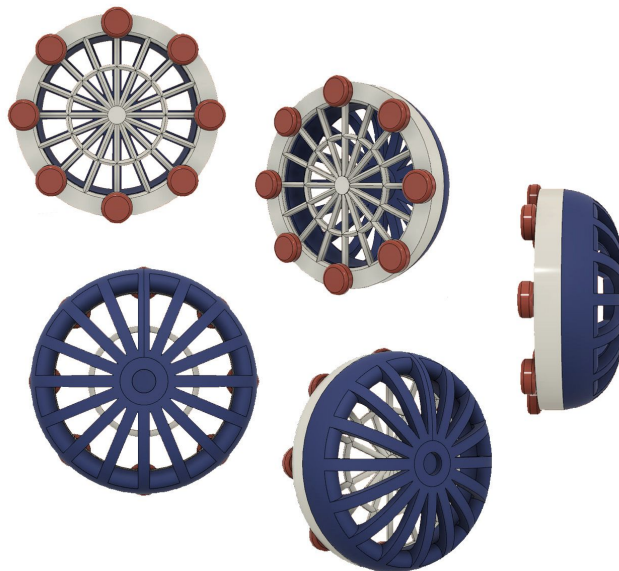
Kiera Gavin, Violet Killy, Cole Legg, Laura Li, Heather Nelson

Final Yoyo Prototype

Our team decided to make a ferris wheel yoyo after considering a spinning-hand clock design, a citrus-inspired design, and a camera aperture design among other ideas. We chose a ferris wheel design because we liked the idea of the yoyo spinning like an actual ferris wheel (albeit much faster) when “yo-ed,” with radial beams spinning around concentric circles as the yoyo revolves around its center axis. Below you can see a sketch of our initial concept.



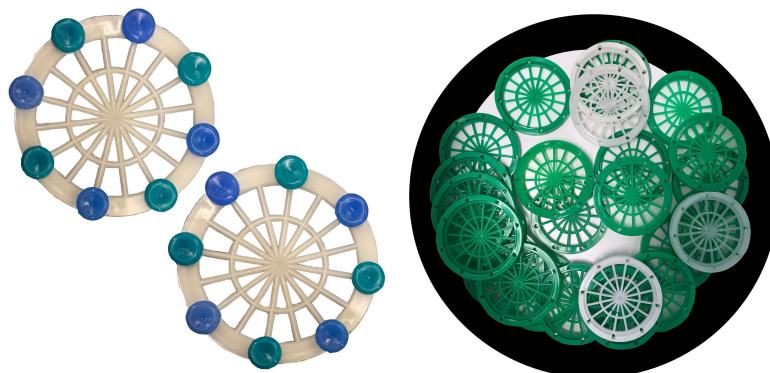
As we developed different ferris wheel designs, we played around with what our 3 injection molded parts and 1 thermoformed part would consist of in the final design. We ultimately decided to have an injection molded snapping depicting ferris wheel beams, an injection molded body part with cutouts to line up with the snapping's beams, injection molded carriages to snap into the snapping, and a thermoformed stand to complete the ferris wheel look. We experimented with the design of a single ring that had a circular pattern of carriages but eventually decided on eight individual carriage pieces to maximize color customization. Below you can see our final design, sans stand.



For our thermoformed part, initial sketches of our design had the ferris wheel sitting on a stand, emulating the real support beams of a ferris wheel. As we continued to design iterations of the yo-yo, we began focusing on a film inside the ferris wheel, on which we would paint a sky-like pattern. As we made our first prototypes of the ferris wheel, however, we liked how the light shone through the yo-yo as if it were a real ferris wheel, so we decided to switch back to our original concept of making our thermoformed part a stand for the yo-yo. The stand is conical, with a strategically shaped top on which the center of the yo-yo can rest. Below, you can see to the left our idea of having an inner painted thermoformed shell within the yo-yo, and to the right, you can see our actual yo-yo sitting upon its thermoformed stand.



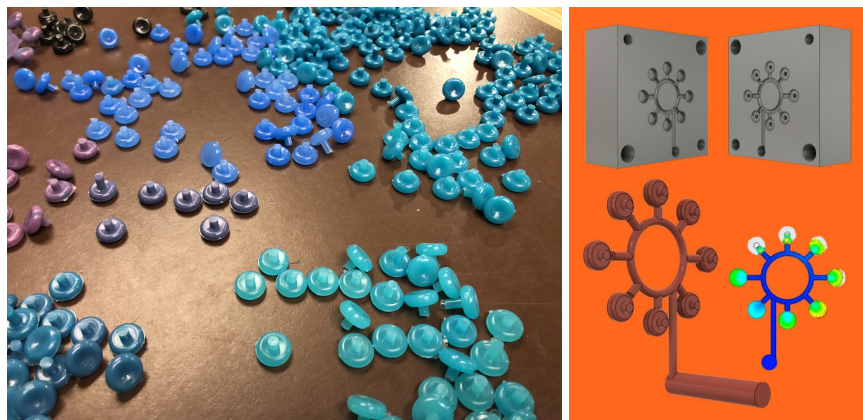
Appearance-wise, the most important aesthetic goal of ours was for our snapping and body beams to be as thin and spindly as possible to best depict a real ferris wheel. On our snapping, the outer-facing sides of the beams have triangular edges to evoke thinness, and the inner-facing sides have rounded edges. We tried to avoid and sharp square edges so that injected plastic would pull away from the mold rather than stick. To overcome concerns about moldflow through the many thin beams, we placed eight gates around the snapping. We strategically aligned the gates with the carriages so that any imperfection from gate removal could be covered with the carriages.



The body part was initially similarly thin and spindly, but moldflow simulations showed that our design was too difficult to injection mold. Thus, we settled on slightly thicker cutouts so that short shots would not be an issue. We only needed four gates for the body since the flow was generally simple. Below, you can observe a moldflow simulation of the updated body (top) and the initial, more complex body, with a short shot (bottom).



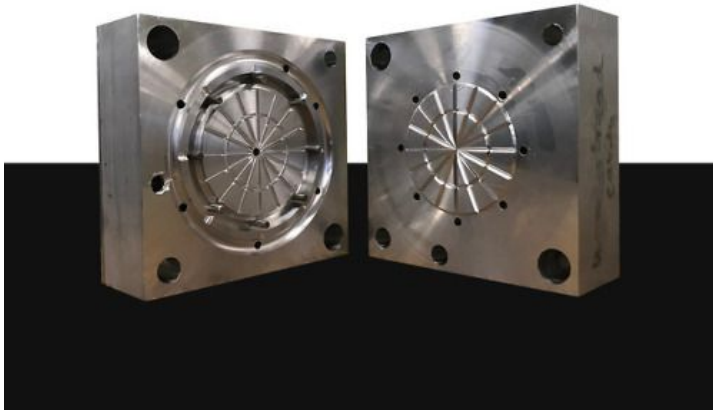
We created the carriage parts using a tree mold where eight carriages aligned with eight ejector pin holes. The core mold had a large runner with small individual gates into each carriage, kept small so that removal wouldn't result in surface flaws.



After settling on this design, we chose to make the snapping of the ferris wheel a light gray color and leave the thermofom stand a clear color. Because we kept the two of them neutral colors, we could be playful with the other parts of the yoyo. We chose bright colors for the body like teal, fuschia, yellow and used the same mixes for carriages as well as blue, light blue, purple, and black. We also experimented with marbling colors for the carriages. Below, you can see our final product - 50 amazing yoyos!



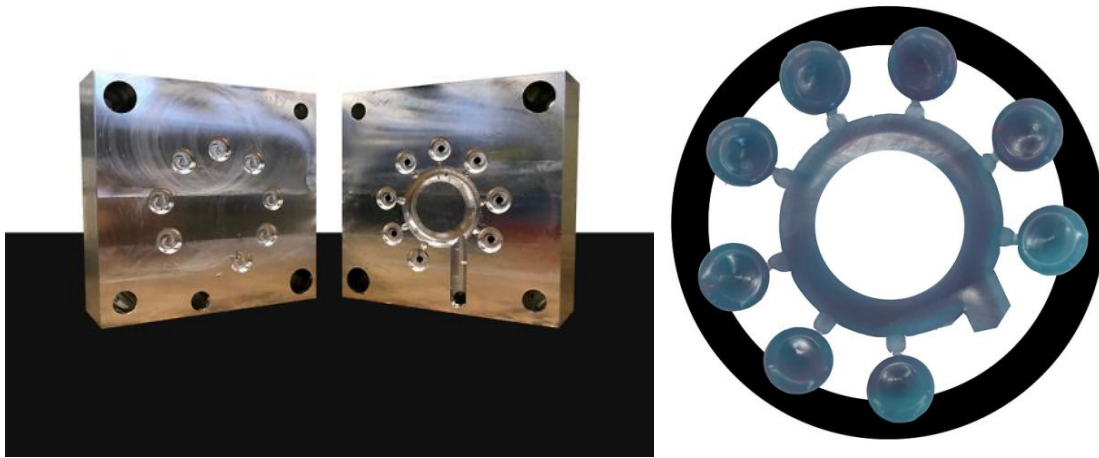
Component 1: Snapping Tooling + Parts



Component 2: Body Tooling + Parts



Component 3: Carriage Tooling + Parts



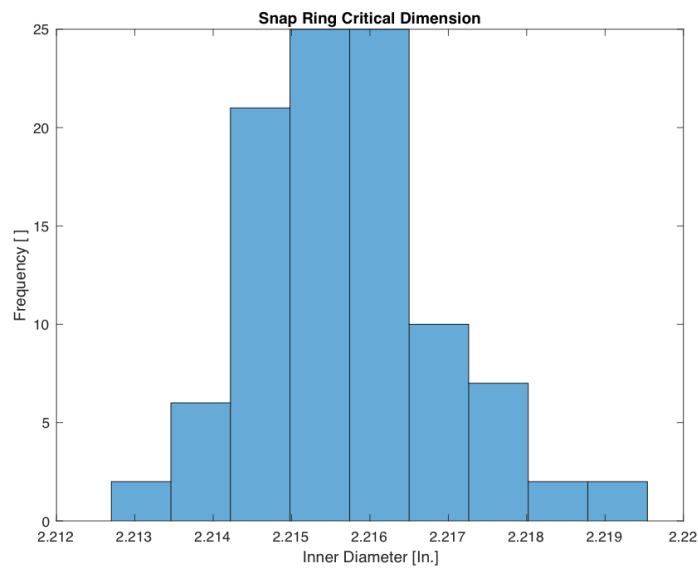
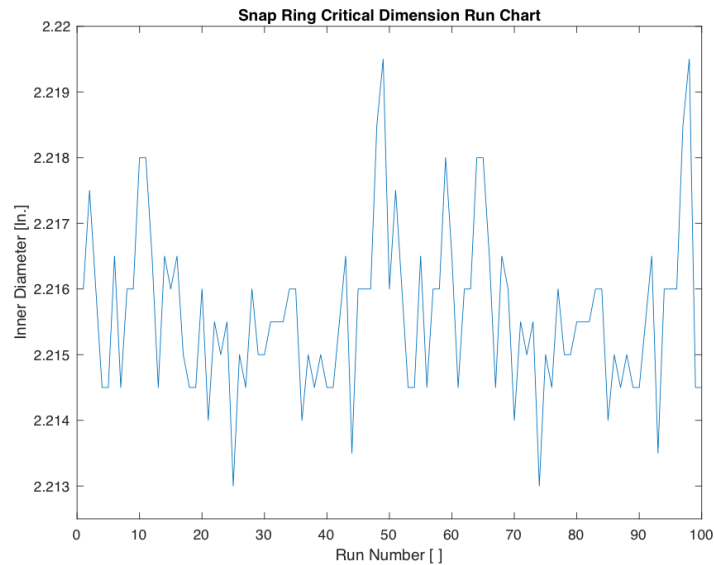
Component 4: Stand Tooling + Parts

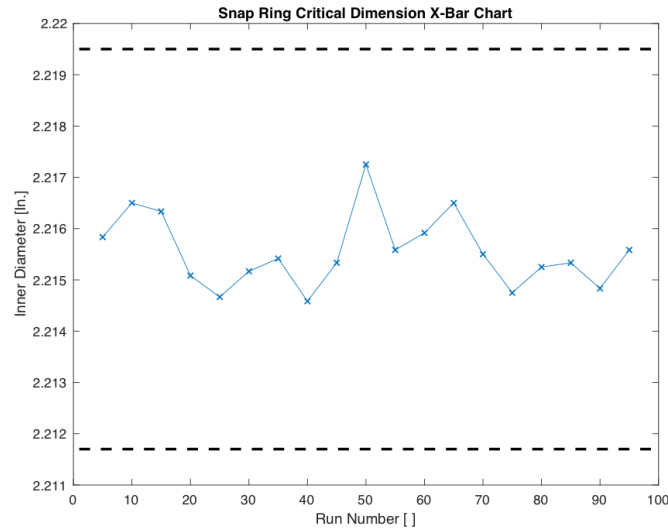


Snapring

Critical Dimension: Inner Diameter

The inner diameter was measured using a set of calipers, and the part was allowed to cool for minimum thirty minutes before any measurements were made.





Overall, our data showed a normal distribution slightly skewed towards higher values. The data had a mean of 2.2156 inches and a standard deviation of .0013 inches.

For the X-Bar chart we chose groups of five yoyos, because, having roughly 100 parts, we felt that any larger groups would not provide enough data points. Alternatively, we felt that smaller groups would not provide meaningful averages (i.e. an average of two parts would be swung too much by a single data point). For our control limits, we used the bounds of three times the standard deviation. This means our upper and lower control limits were 2.2195 and 2.2117 inches respectively. These control limits were selected based on the fact that we would expect a 99.7% yield rate from our parts with these control limits, given that the data is distributed normally.

On the X-Bar chart we can see that our averages stay well within our upper and lower control limits. One can notice some disturbances, for example a spike in the data around point number 50. This is because part number 49 was measured to have the maximum inner diameter of 2.2195 inches. This outlier could be seen as the cause, and may be due to human inaccuracies when measuring or perhaps the piece not fully cooling/shrinking before the measurement was taken.

Process Capability for Process

Our results and limits lead to Cp and Cpk values of 1. Cp is defined as the ratio of the engineering tolerance to the natural tolerance. Defining engineering tolerance as the difference between our upper and lower control limits (6 standard deviations) and natural tolerance as 6 standard deviations, we obtain a Cp value of 1. Cpk is defined as the minimum of difference between the average and the upper and lower control limits, divided by half of the natural tolerance. Because our control limits were centered around our mean value, we obtain a Cpk value equivalent to our Cp value of 1.

In typical production standards you may see Cp and Cpk values of 1.34, which are slightly higher than our own, but a Cp value of 1 still indicates that we can expect a yield rate of roughly 95%.

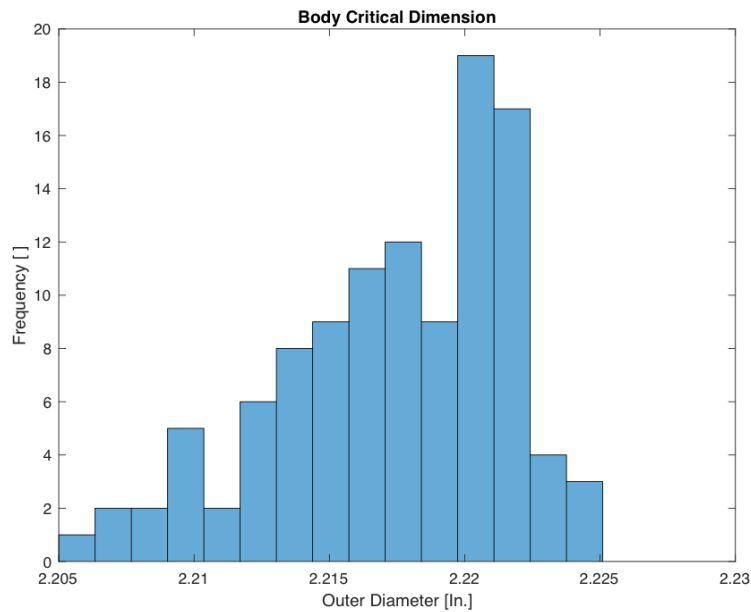
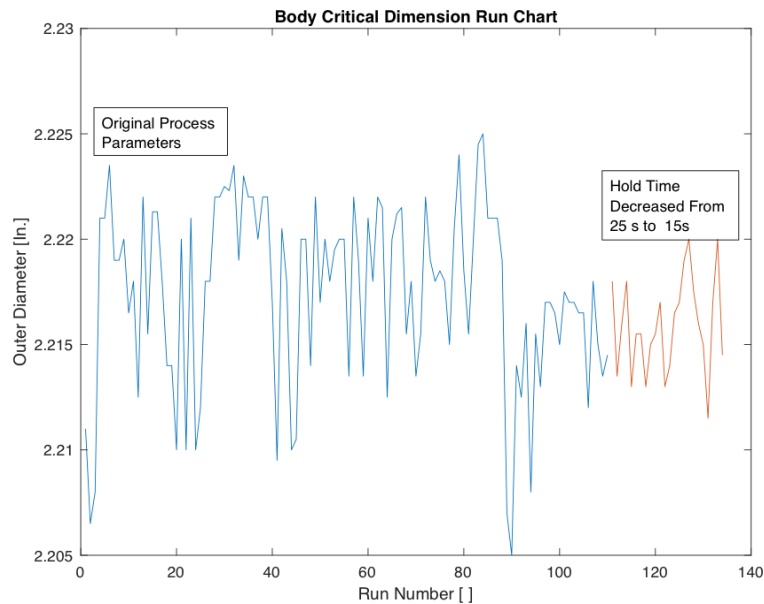
Source:

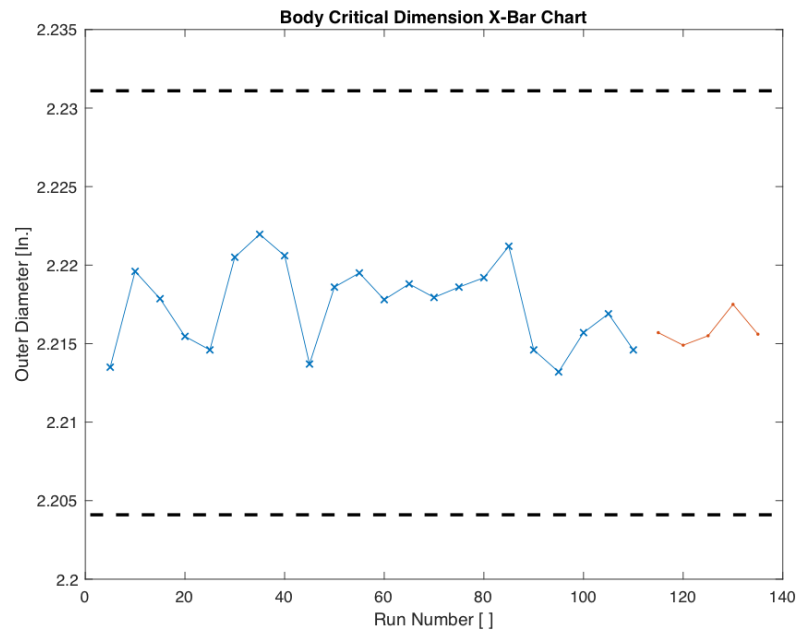
<http://blog.minitab.com/blog/statistics-and-quality-improvement/process-capability-statistics-cp-and-cpk-working-together>

Body

Critical Dimension: Outer Diameter of Lip

The outer diameter was measured using a set of calipers, and the part was allowed to cool for a minimum of thirty minutes before any measurements were made.





Overall, our data showed a normal distribution slightly skewed to lower values. The data had a mean of 2.2176 inches and a standard deviation of .0045 inches.

For the X-Bar chart we chose groups of five parts for the same reasons brought up in the snapping section. For our control limits, we used the bounds of three times the standard deviation. This means our upper and lower control limits were 2.2311 and 2.2041 inches respectively.

On the X-Bar chart we can see that our averages stay well within our upper and lower control limits, even for our extra 25 bodies which had a decreased hold time of 15 seconds as opposed to the 25 seconds we used on our first 110 bodies. One can notice some disturbances, for example a large drop between run numbers 80 and 90. This steep drop is probably due to data points 89 and 90: the two lowest data points of our entire runs. These diameters came in at 2.2070 and 2.2050 inches respectively. These low numbers could be explained by a human error inputting the measurements or just rare natural variance to some very low values in our distribution.

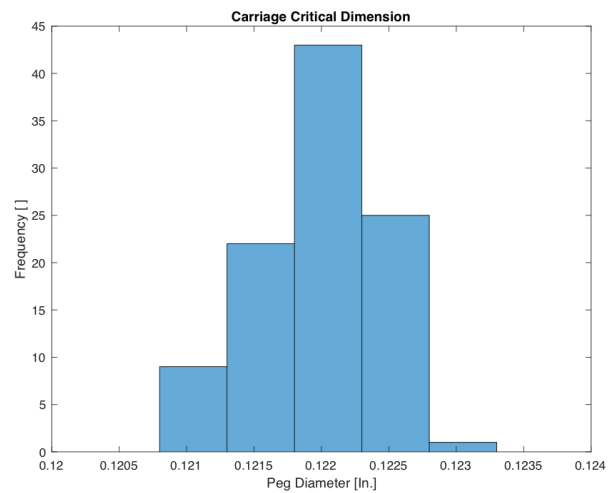
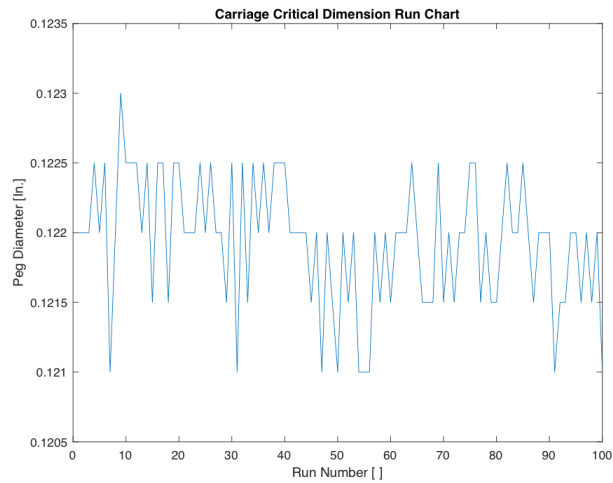
Process Capability for Process

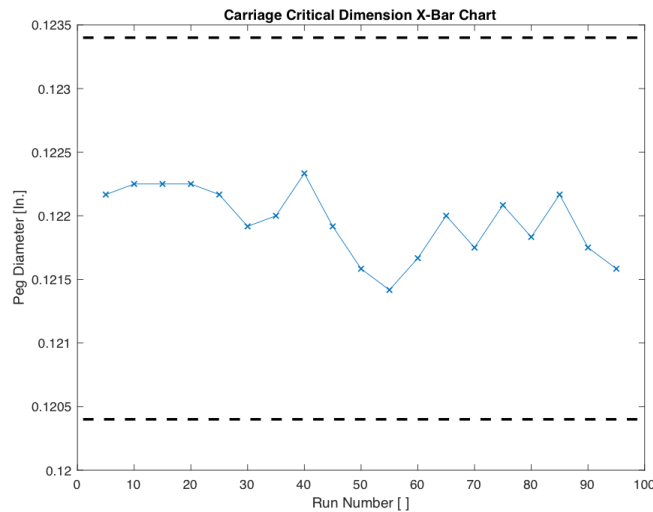
Our calculations for C_p and C_{pk} are independent of our parts because we set our control limits based off of the standard deviation of our parts. This means that the C_p and C_{pk} values for the body were also 1.

Carriages

Critical Dimension: Peg Diameter

The peg diameter was measured using a set of calipers, and the part was allowed to cool for a minimum of thirty minutes before any measurements were made.





Overall, our data showed a normal distribution with a steep curve. The data had a mean of 0.1219 inches and a standard deviation of 0.0005 inches. We believe that the reason for the steep curve is because of a combination of the low standard deviation as well as the small number of bins in the histogram. Due to the accuracy of our calipers, there were only five discrete values for the diameters measured, therefore, no more than five bins in the histogram would have made sense.

For the X-Bar chart we chose groups of five parts, for the same reasons as previously stated. For our control limits, we, once again, used the bounds of three times the standard deviation. This means our upper and lower control limits were 0.1234 and 0.1204 inches respectively. These control limits were selected for the same reasons as in previous sections.

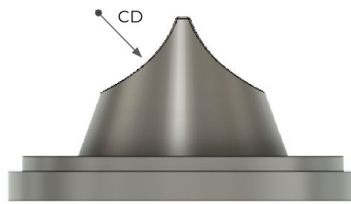
On the X-Bar chart we can see that our averages stay well within our upper and lower control limits, and there are no significant disturbances on the chart.

Process Capability for Process

Our calculations for Cp and Cpk are independent of our parts because we set our control limits based off of the standard deviation of our parts. This means that the Cp and Cpk values for the carriages were once again 1.

Thermoformed Stand

After consideration on our thermoformed part, we decided that our stand did not necessarily have a critical dimension. This is because the stand's functionality is only dependent on a yoyo's ability to rest on top of it, and this is a simple task that would still be functional even with relatively large fluctuations in dimensions. If we were to choose a critical dimension, it would be the critical dimension pictured below.

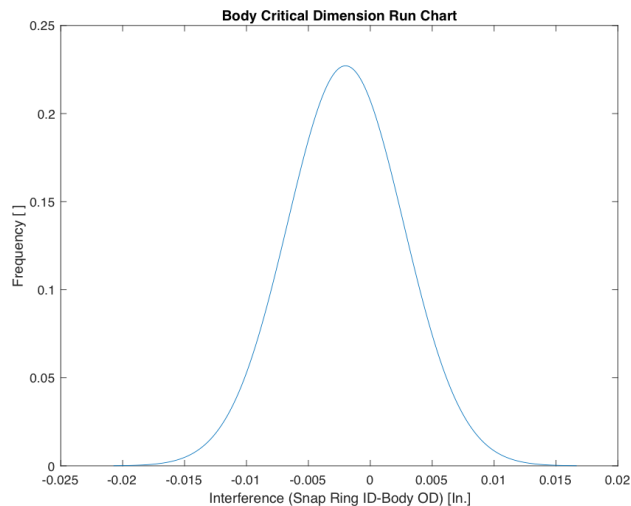


However, we believe that the best way to assess the quality of our thermoform stand was not to try to measure this radius. Instead, we tested the quality of our thermoformed stand simply by placing a yoyo on top of our stand and seeing if it functioned. We found a 100% success rate with our stands.

If we were to manufacture these stands at a larger scale we would recommend testing quality by placing a sample, such as a 3D printed yoyo or a yoyo that has had its critical dimensions measured, on top of the stand.

Because we had a 100% yield rate, and we expect very large tolerances, we can assume our C_p and C_{pk} values would be very large and possibly exceed industry standards. We believe that C_p and C_{pk} values around 2 are reasonable.

Final Tolerances to Critical Dimensions



The graphic above represents the normal distribution of the snapping inner diameter minus the body outer diameter. This is a good gauge for the snap fit between the two. This gives us an the difference between the diameters average -.002 inches. This means that the interference between the parts is roughly two thousandths of an inch on average. We also find a a standard deviation of 0.0047". For our calculations, we assumed that an acceptable snap fit happens with an difference of .001" to -.005".

In order to make 99% of our parts fall between this range keeping the current average values for our body OD (2.2176") and snap ring ID (2.2156"), the interference standard deviation must then be .001". This value is obtained from the fact that a difference from the mean of three times the standard deviation of .001" would place us at our control limits of .001 and -.005".

If interference standard deviation is .001" and the snapping and body have equal standard deviations, then each of their standard deviations must be $.001/\sqrt{2}$.. This means the standard deviation for the body and the snapping each part must be .0007".

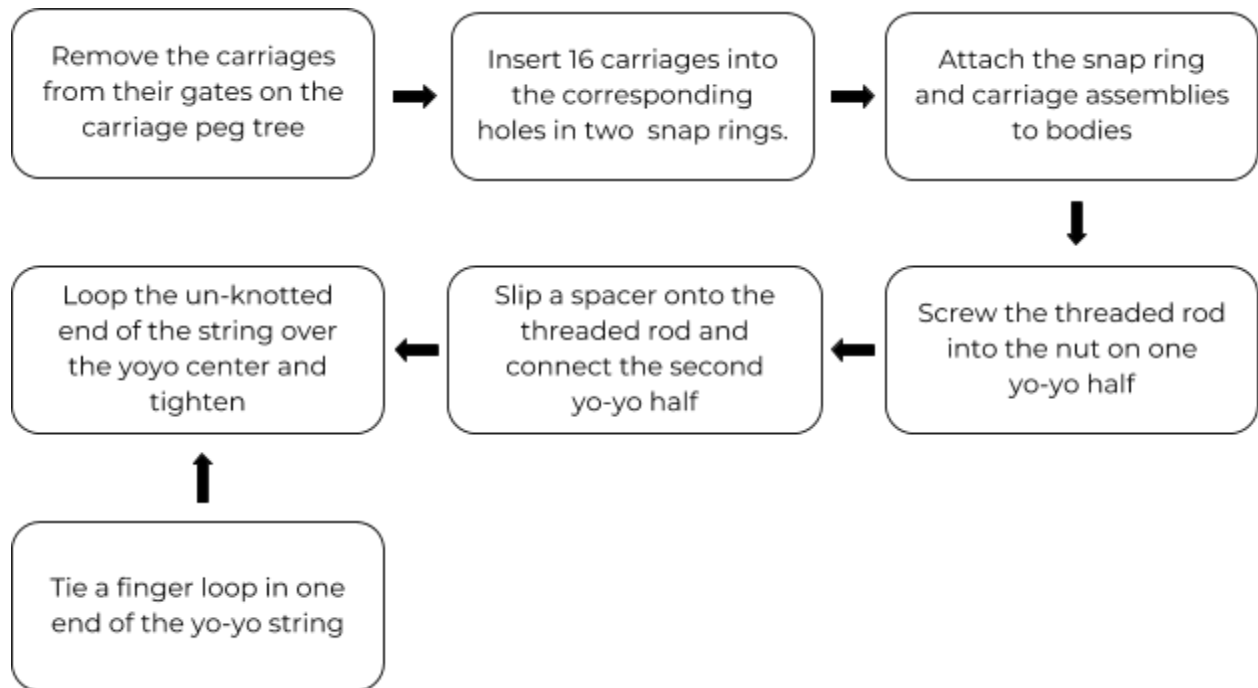
Because we want a 99% yield rate, we should use the values three standard deviations away from the mean as our upper and lower control limits. If we were to move into mass production, our final tolerances would be 2.2155" and 2.2197" for our body, 2.2135" and 2.2177" for our snap ring, and .1204" and .1234" for our carriages.

Given that during our assembly process we made 800 carriages and put them all into snap rings with a failure rate of 0%, we feel that a sample size that large proves that our current tolerances for the carriages are satisfactory. For this reason we carried over the mean of .1219" and the standard deviation

of .0005” from our production run and would apply this same standard to a larger scale production.

Part	-3σ	-2σ	-1σ	0	1σ	2σ	3σ
Body OD (inches)	2.2155	2.2162	2.2169	2.2176	2.2183	2.2190	2.2197
SR ID (inches)	2.2135	2.2142	2.2149	2.2156	2.2163	2.2170	2.2177
Carriage Peg Diameter (inches)	.1204	.1209	.1214	.1219	.1224	.1229	.1234

Assembly Diagram



Step	Estimated Cycle Time
Remove the carriages from their gates on the carriage peg tree	60 seconds
Insert 16 carriages into the corresponding holes in two snap rings.	100 seconds
Attach the snap ring and carriage assemblies to bodies	6 seconds
Screw the threaded rod into the nut on one yo-yo half	9 seconds
Slip a spacer onto the threaded rod and connect the second yo-yo half	8 second
Tie a finger loop in one end of the yo-yo string	9 seconds
Loop the un-knotted end of the string over the yoyo center and tighten	11 seconds
Total:	203 seconds or 3.38 minutes

The best option for optimizing the line would be to parallelize a number of these steps. For instance, we had one person who focused solely on detaching all of the carriages from the carriage trees and sorting them by color. We had two members who would each assemble one yo-yo half, so then both halves could be passed off at the same time to be assembled together.

Robotics and Automation

During the manufacturing process, our yoyo production rate would have increased if the injection molding process were automatic as opposed to semi-automatic. Fewer man-hours would be necessary for production runs. Unfortunately, with our current molds, using the automatic setting would not have been possible. For some of our parts, we had problems with the plastic getting stuck in the mold. For our snaprings, this was consistently a problem — when we used mold release, our part would flash for a couple of cycles, and then it would be useable for a few cycles, and then after about 10 cycles, the part would get stuck. So, using this automated manufacturing technique would require a small redesign of the mold to optimize ejection.

Our assembly process could have been greatly streamlined by the use of robotics. The most time intensive step in our assembly line is the de-treeing and placing of the carriages. Our snap fit for the carriage pegs into the holes in the snap ring is perfect — while pressing the pegs in, one can hear a satisfying ‘snap,’ after which the carriage cannot be removed without pliers. Because of how precise our snap fit is, it requires a thumb to apply a considerable amount of force onto the carriage to press it in. When doing this hundreds of times, it not only takes time but is somewhat tiring to do one by one. If we had a robot arm that could hold onto the whole carriage tree and apply force onto each of the individual carriages to press them all at once into a snap ring, our entire assembly process could shorten by up to two minutes. This arm would only need to apply a downward force onto a jig that we would need to create to hold the carriage tree and to direct the downward force to each of the carriages. To achieve this concept as a whole, we would need to redesign our carriage mold so that each of the carriages is injection molded to be the proper spacing away from each other to fit into the snap ring without de-treeing all of them. This might also require a redesign of the snapring to accommodate holes to align with the carriages, since the carriage placement is currently constrained by location of ejector pins (the pegs of the carriages are comprised of plastic filling into ejector pin holes up to a certain length, which is then blocked off by the ejector pin).

The rest of our assembly process isn't as time intensive and is easier to do by hand. Unless we were to increase our desired production *significantly* (to many thousands, millions of yoyos), we do not recommend automating more of the assembly process. For our yoyo to meet our aesthetic standards, we expect the geometry of the snap ring to line up with the geometry of the body part, and the other half of the yoyo. While snapping the snap ring onto the body and screwing it on to another half would potentially not be difficult to automate, it would be much more difficult create an automation process precise enough with perfect registration to guarantee the geometry we are looking for in a finished yoyo. The human eye and hand is much better at doing that, and the whole process only adds up 43 seconds per yoyo, including tying and wrapping the string, which is acceptable at our volume and high volume production up to a certain point.

AM Processes and Materials

Service Bureau 1: **Shapeways**

Material: Versatile plastic

- We chose this option for all of our yoyo parts because it was the cheapest material that fit our needs. It is a nylon plastic used for a variety of applications, 'ranging from functional to decorative products.' It is advertised as strong, durable, skin-safe, and supporting of thin features, all key characteristics of our yoyo.

Finish: Processed

- This finish removes some material to create a smoother surface.
- We used this on the body and snap ring because they comprise the biggest parts of the yoyo and having a nice surface finish on the snap-fit is imperative for assembly

	# Parts/ Yoyo	Total cost/part for 1 yoyo	Total cost/part for 50 yoyos	Material	Finish	Color
Body	2	\$20.40	\$20.40*	Versatile plastics	Processed (+\$1)	Pink (+\$3.50)
Snapring	2	\$8.93	\$8.93*	Versatile plastics	Processed (+\$1)	White
Carriage	16	\$10.35	\$10.35*	Versatile plastics	None	Blue (+\$3.50)
Stand	1	\$12.47	\$12.47*	Versatile plastics	None	White

*The price didn't change on Shapeways when changing the quantity from 2 to 100.

We then decided to check a different service bureau to compare prices, materials, and other options because Shapeways did not offer as many options as we were expecting.

Service Bureau 1: **Materialise**

Method: Stereolithography

- We chose stereolithography because it produces finer quality parts and our yoyo assembly requires precision parts for the snap fits to work properly. ProtoGen White was the cheapest material that seemed to meet the requirements for the durability for a yoyo. Other materials had

higher performance quality, but we decided that properties like water resistance and high elongation at break were not imperative to our design.

Material: ProtoGen White

- We chose this material because it was the cheapest material that fit our yoyo application. It is listed as being suitable for general purpose applications, resilient with good surface quality and thermal properties, having a short lead time of 3 working days, and having maximum part dimensions greater than our parts' dimensions.

Finish: Basic finish

- We decided to use the basic finish because when we printed our parts on the Form2 in LMP, our yoyo fit together without needing any form of sanding. For the purposes of this report, we did not add any color to our yoyos because this is for a prototype model.

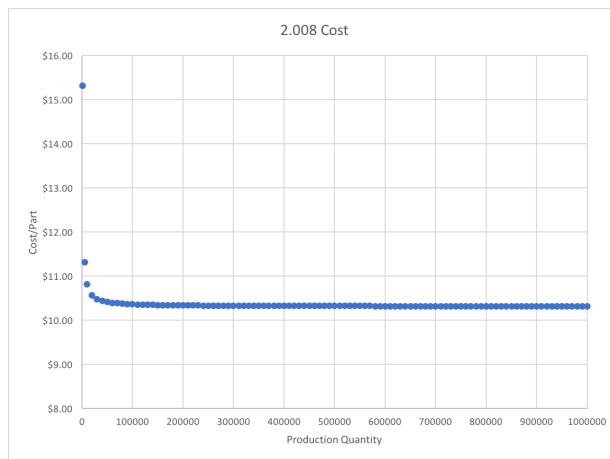
	# Parts/yoyo	Total cost/part for 1 yoyo	Total cost/part for 50 yoyos	Material	Finish
Body	2	\$97.92	\$56.10	ProtoGen White	Basic
Snapping	2	\$100.98	\$51.00	ProtoGen White	Basic
Carriage	16	\$27.54	\$26.52	ProtoGen White	Basic
Stand	1	\$110.16	\$51.00	ProtoGen White	Basic

Per-Unit Cost of Yoyo Manufacturing

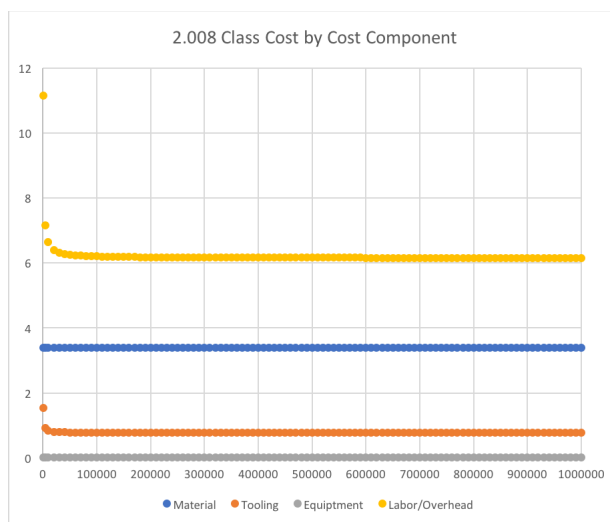
The assumptions and methods for calculating fixed and variable costs for each yo-yo are outlined in [this spreadsheet](#).

Case 1: Prototype using 2.008 Class Process

The 2.008 Yo yo process cost approaches \$10.32 as production scale increased significantly.

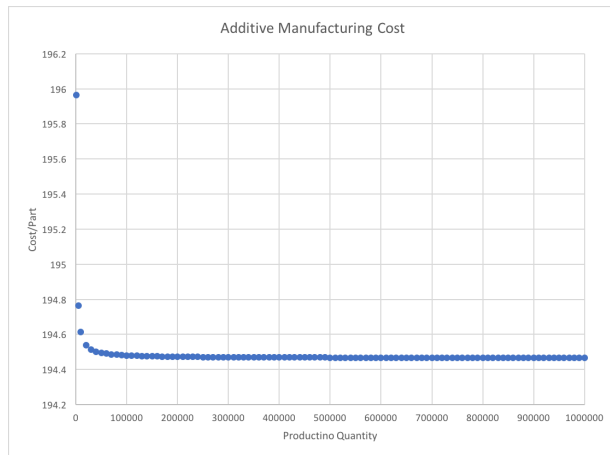


We see that as production increases, the most impactful cost components that are changing are Labor/Overhead (yellow below) and Tooling (red below)



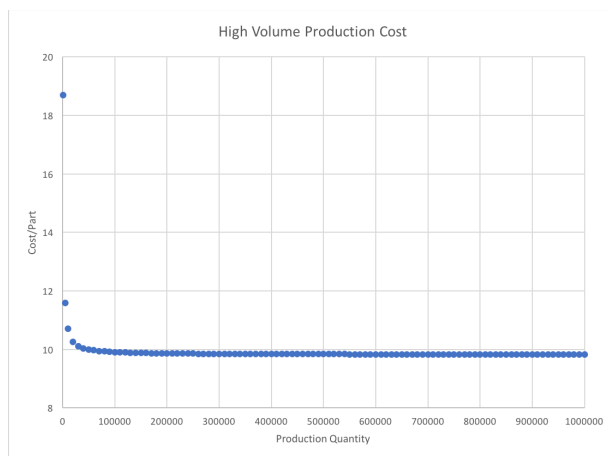
Case 2: Additive Manufacturing Process

Based on our quotations, attempting to produce these parts using additive manufacturing is not currently realistic. In addition to cost, the capability and capacity of additive manufacturing is not yet scalable to reasonably produce the demand we are looking for of about 1 million Yo-yos.

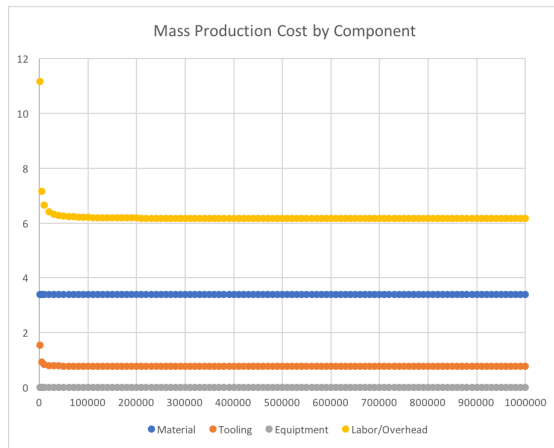


Case 3: High Volume Production Process

The cost for our mass produced Yo-yo begins slightly higher than our 2.008 Class Yo-yo based on the investment in tooling. We see the costs approach \$9.83/part as the production increases.



Similarly to our 2.008 Prototype making process, we see the Labor/Overhead costs and the tooling costs decrease most significantly as volume increases.



2.008 Cost Components compared to Mass Production Cost Components

Process	Part Count	Material	Tooling	Equipment	Labor/Overhead	Total [\$/part]
2.008	50	3.38	15.36	0.007	106.15	124.90
2.008	5000	3.38	0.92	0.007	7.15	11.31
2.008	100000	3.38	0.78	0.007	6.20	10.37
Mass Production	50	3.38	112.56	0.007	71.43	187.38
Mass Production	5000	3.38	1.13	0.007	7.08	11.59
Mass Production	100000	3.38	0.06	0.007	6.46	9.91

We broke down the two different production options by component to compare cost across different volumes. We found the crossover point occurs near the 5000 part mark.

A key takeaway from this analysis is that the tooling cost and Labor/Overhead costs are the biggest factors in contributing to cost at low scale production, but as production quantity grows, the material cost plays a more significant role in the overall cost.

Crossover Point where Additive Manufacturing is more cost effective than the 2.008 Process

We found that there is no such point where it would be cost effective to use an additive manufacturing process over our in-class production model. This is likely due to the high cost of our individual component parts for AM. The different parts are very intricate and our quotes are priced relatively high.

Comparison: Final Production YoYo vs. Purchased Sample Yoyo

The Duncan Yoyo purchased from Amazon costs \$4.71 (source: https://www.amazon.com/Duncan-Imperial-Assorted-colors-Pack/dp/B000H6DY5U/ref=sr_1_5?ie=UTF8&qid=1544480179&sr=8-5&keywords=duncan+yoyo).

When mass produced at 1,000,000 units in the 2.008 lab, our yoyo costs \$10.32, so our yoyo is not competitive cost-wise with the commercially produced one.

Looking at the commercially produced yoyo we see one gate defect on each half of the yoyo. These gate defects are larger than the defects on the outside of our yoyo; however, we had four gates on our yoyo. Another visible defect is some scratched off paint on the face of the yoyo, and frayed ends of the yoyo string. The parting line of the injection mold is also visible each side, occurring at the widest part of the yoyo.

We believe that the two other attributes that would most be important to a customer would be the aesthetics of the yoyo and the “yo-ability.” In a head-to-head competition we believe that we split these categories with the commercially produced yoyo. Our design is, of course, the more elegant and aesthetic yoyo. The commercially produced yoyo, having a bit more weight to it than our own, has a smoother yo to it.

For all of the reasons listed above, we believe the commercial yoyo would appeal to a market of hardcore yo-ers who want a yoyo that yos well, and does it for a low price. Our yoyo appeals to a younger generation of yo-ers, with more of a hipster mentality. Our customers are willing to pay a premium price for an aesthetic yoyo that will make all of their friends jealous.